Mohammad Rahmanitalab

B.Sc. Mechanical Engineering | Minor in Computer Science | Sharif University of Technology

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LinkedIn | GitHub

Education

B.Sc. in Mechanical Engineering, Sharif University of Technology | Tehran, Iran *2021 – 2026*

Thesis Supervisor: Prof. Saeed Behzadipour

- **B.Sc. Thesis:** Machine Learning-Based Motion Analysis for Duchenne Muscular Dystrophy Rehabilitation.
- GPA: **3.95/4.00** (18.94/20) — Ranked **12th out of 116** students in the department.

B.Sc. Minor in Computer Science, Sharif University of Technology | Tehran, Iran *2022 – 2026*

- GPA: **4.00/4.00** (18.98/20).

Research Interests

- **Machine Learning:** Time-series and sensor-data learning, supervised learning, representation learning
- **Biomedical AI & Motion Analysis:** Wearable sensing, IMU-based human-motion analysis, biomechanics, movement classification and clinical applications
- **Robotics & Control:** Medical and rehabilitation robotics, haptics, teleoperation, robot modeling and control

Research Experience

Machine Learning-Based Motion Analysis | Sharif University of Technology 2024 – 2026

- Led an end-to-end IMU-based human-motion analysis project for Duchenne muscular dystrophy (DMD) rehabilitation, covering data collection, preprocessing, feature extraction, and supervised movement classification.
- Developed the analysis pipeline in Python and MATLAB, including signal preprocessing, feature extraction, supervised classification, and analysis of activity-specific motion patterns relevant to quantitative rehabilitation assessment.
- Conducted the research under the supervision of Prof. Saeed Behzadipour and prepared the study for dissemination as an ICBME manuscript.

Research Output

Manuscript in Preparation | ICBME 2026

- **M. Rahmanitalab**, S. Behzadipour, and M. Babaee. "An Activity-Specific Machine Learning Framework for Classifying Simulated NSAA Performance Levels Using Wearable IMUs." Manuscript in preparation for submission to ICBME 2026.

Technical Skills

Programming: Python, Java, C, Go
Development: Git, Linux
Machine Learning & Data Analysis: NumPy, pandas, SciPy, scikit-learn; feature engineering, supervised classification
Scientific Computing: MATLAB, Simulink
Robotics & Control: ROS 2, Gazebo, TwinCAT 3, MSC ADAMS; kinematics, system modeling, PID control
CAD: SolidWorks
Languages: Persian (Native), English (Professional working proficiency)

Professional Experience

Back-End Developer | Moj Gostaran Adak, Tehran Oct – Dec 2025

- Developed Java backend services integrated with the Traccar GPS platform for real-time device communication and data handling.
- Implemented Go backend services and RESTful APIs for an internal locker-tracking platform deployed in the Hamrah Aval project.

Robotics Experience

Robotics Engineering Intern | Sina Robotics & Medical Innovators Co., Tehran Jun – Sep 2024

- Derived kinematic solutions and simulated components of master–slave surgical robotic systems using MATLAB and MSC ADAMS.
- Worked on haptic teleoperation and real-time robotic-system development using TwinCAT 3, with emphasis on precision and responsiveness.

Selected Technical Projects

Programming

Networked Multiplayer Java Application Spring 2025

Developed an online multiplayer application using object-oriented design, multithreading, socket-based network programming, and client–server architecture.

Robotics and Control

Multi-DOF Robotic Manipulator Modeling & Simulation Fall 2024

Modeled and simulated a multi-DOF robotic manipulator in MATLAB/Simulink, studying forward/inverse kinematics and dynamic behavior.

Control & Dynamic Systems Projects 2024

Designed and analyzed PID-controlled and dynamic systems including ball-and-beam and rotating-governor systems, and simulated a pushing mechanism in MATLAB/Simulink.

Teaching Assistant Experience

Fundamentals of Automatic Control Design | Sharif University of Technology Fall 2025

- Supported undergraduate instruction, assessment, and student guidance in controller design, PID tuning, and stability analysis.

Automatic Control | Sharif University of Technology Spring 2025

- Supported undergraduate instruction, assessment, and student guidance in control theory, feedback systems, and Lyapunov stability analysis.

Academic Honors & Distinctions

National Master's Entrance Examination | Iran 2025

- Ranked **85th** nationally in Iran's Master's Entrance Examination.

Direct M.Sc. Admission | Sharif University of Technology 2025 & 2026

- Qualified for direct M.Sc. admission based on academic excellence.

Selected Coursework

Robotics & Control:

- Robotics
- Automatic Control
- Fundamentals of Automatic Control Design
- Measurement & Control Systems
- Dynamics of Machinery

Computer Science & Mathematics:

- Advanced Programming
- Operating Systems
- Data Transmission & Networks
- Linear Algebra
- Probability
- Computer Simulation

Leadership & Service

CARE Student Society: Media Director; managed media content, production, and team operations.

ICERS 2024 & Dept. Website: Contributed to IT/media coordination and developed/maintained the Mechanical Engineering Department website (WordPress).